

Silicon Core Harvesting is Visible in Supercomputer Telemetry: A Fleet-Scale Study of 1,960 POWER9 Sockets

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Abstract—Processor vendors sell partially defective dies by permanently disabling faulty units, a practice known as core harvesting. The resulting per-die disable maps are proprietary and are conventionally invisible to system owners. We show that the out-of-band (BMC/IPMI) sensor telemetry routinely collected by production supercomputers exposes these maps directly, and we recover them for all 1,960 processor sockets of CINECA’s Marconi100 from the public M100 ExaData release [1]. Every socket exposes exactly 16 of 24 possible per-core thermal sensors, and the eight absent sensors always form four complete *slice* pairs $(2k, 2k+1)$ —the POWER9 unit that shares an L2/L3 block [2], [3]—with zero exceptions. We observe 442 of the 495 combinatorially possible patterns, establishing that the map is per-die rather than a fixed SKU. Using a curveball null that preserves both per-socket and per-slice marginals, we show harvested slices cluster in sensor-index space ($z = -20.8$). Thermal analysis confirms that the two cores of a slice are physically abutted (residual correlation $r = +0.51$ within-slice versus $+0.29$ for cross-slice neighbours at identical index distance), but we find only weak evidence that the slice index itself is monotonic in physical die position ($p = 0.025$; a 3×4 grid fits the thermal coupling better than a linear arrangement). We detect a sharp procurement-lot boundary at rack 22 (Welch $t = 27.3$) separating two populations with markedly different harvesting statistics. Finally, tracking every socket across the full 2.5-year, 1.67-million-socket-day record, we find harvest maps behave as near-static manufacturing constants: 89.5% of sockets never change configuration, and none of the 235 observed transitions occurs during operation—every one coincides with a reboot. Against that backdrop we isolate three episodes, on three of 1,968 sockets, in which a socket ran on 14 cores for 2–4 days, each losing exactly one slice and at least one subsequently restored: to our knowledge the first direct observation of POWER9 field core-guarding in a production system, at a measured rate of 0.15% of sockets over 2.5 years.

Index Terms—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance

I. INTRODUCTION

Every large processor die ships with defects. Rather than discard them, vendors permanently fuse off the faulty units and sell the remainder as a lower-core-count part—*core harvesting*. The practice is universal and openly acknowledged, but the resulting *per-die* maps of which units were disabled are proprietary: a buyer is told the part has 16 cores, never *which* 16. Those maps encode information about a fab’s defect process, a vendor’s binning policy and a procurement’s provenance that is otherwise unavailable outside the manufacturer.

Independently, HPC operators have begun publishing holistic out-of-band monitoring at fleet scale [1]. We observe that such telemetry carries an unintended side channel: a baseboard management controller exposes one thermal sensor per *physical* core position and does not renumber the surviving cores compactly, so the set of sensors that report *is* the die’s harvest map, published inadvertently and at fleet scale.

We exploit this to recover harvest maps for every processor socket of CINECA’s Marconi100, a Top-10 system, from a public dataset, and to track them across 858 days. Our contributions are:

- A recovery method (Sec. III) with an explicit validity argument, needing only per-core sensor telemetry that operators already collect.
- The first fleet-scale characterisation of core harvesting in a deployed supercomputer: the granularity is exactly the POWER9 slice, maps are per-die rather than per-SKU, and harvested slices cluster in sensor-index space.
- Evidence of *procurement-lot structure*—a sharp change-point separating two populations with markedly different binning statistics—from telemetry alone.
- A longitudinal result over 1.67 M socket-days: maps are static, configuration changes occur only at reboot, and three field core-guard episodes are isolated and corroborated.
- A negative result that matters operationally: the harvest map has no practical effect on socket power (Sec. IV-I), so schedulers need not model it.

II. BACKGROUND

A. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a 695 mm^2 die in 14 nm FinFET with 17 metal layers [3]. Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a slice, and each slice owns 512 kB of L2 and 10 MB of L3* [3], [2]. IBM’s Hot Chips 28 presentation shows the 24 cores above $12 \times 10\text{ MB}$ L3 blocks, attached to a 7 TB/s on-chip fabric at $256\text{ GB/s} \times 12$ [2]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with $2 \times$ 16-core POWER9 at 2.6/3.1 GHz and $4 \times$ NVIDIA V100 [4], [5]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 1 shows the organisation annotated with the harvest rates we measure.

B. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, “Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23” [6]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. METHOD

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s, k] \in \{0, 1\}^{12}$ over slices $k = \lfloor \text{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \dots, 23\}$, so the index is a fixed physical-position identifier.

B. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [7], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. Row- and column-sum preservation is asserted on each of 200 draws.

C. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. RESULTS

A. Every socket is 16 cores; granularity is the slice

Populations: Three socket populations recur and are kept distinct throughout: $\mathcal{S}_{\text{all}}$, the 1,968 sockets appearing anywhere in the record (984 nodes); $\mathcal{S}_{\text{map}}$, the 1,962 with at least one clean day and hence a well-defined harvest map; and $\mathcal{S}_{2004}$,

TABLE I
DISABLE-MAP STRUCTURE OVER THE FULL 31-MONTH RECORD.

Sockets with a well-defined harvest map	1,962
Median clean days supporting each map	775
Socket-days reporting exactly 16 cores	99.985%
Sockets ever reporting 24 cores	0
Harvested cores form complete $(2k, 2k+1)$ pairs	1,962 (100%)

the 1,960 present in the single month 2020-04. All structural results below are over $\mathcal{S}_{\text{map}}$ and the full 31-month record.

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

B. The map is per-die, not per-SKU

443 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 7). The most common (slices 0–3) covers only 5.5% of sockets. Within a node the two sockets match in just 1.2% of cases, with mean overlap 1.648 slices against a shuffled null of 1.474 ± 0.026 ($z = +6.6$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.2% of sockets against a uniform expectation of 33.3% ($\chi^2 = 807.2$, $\text{df} = 11$, $p \ll 0.001$); Fig. 2 splits this by lot. These marginals are remarkably stable in time: across the 31 months the largest standard deviation of any slice’s monthly rate is **0.25 percentage points**, and slice 0 ranges only 60.9–61.8% (Fig. 8). The single-month snapshot used in earlier drafts was therefore representative, and every estimate here is a 2.5-year-stable measurement.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.242 versus 4.582 ± 0.017 ($z = -20.2$). Applying a Bonferroni correction across all $\binom{12}{2} = 66$ pair tests ($|z| > 3.83$), 17 pairs remain significant (Fig. 3). The positive excesses are 0&1 ($z = +16.7$), 6&7 (+9.2), 2&3 (+8.0), 8&9 (+7.7), 10&11 (+7.5), 4&5 (+7.1), 1&2 (+6.9) and 0&2 (+6.3): **eight of the nine positive-excess pairs have $|\Delta k| \leq 2$** , and every significant *deficit* is a distant pair. We report the single exception rather than rounding it away.

E. Slice siblings are physically abutted

Fig. 4 shows residual correlation against core index distance. At *identical* distance of 1, within-slice siblings reach $r = +0.51$ versus +0.29 for cross-slice neighbours; per-node the contrast between near (≤ 2) and far (≥ 12) pairs is $+0.45 \pm 0.20$ with 40/40 nodes positive (paired $t = +13.9$). The two cores of a slice are roughly twice as coupled as equally-numbered cores across a slice boundary, confirming both physical abutment and the slice as a physical unit.

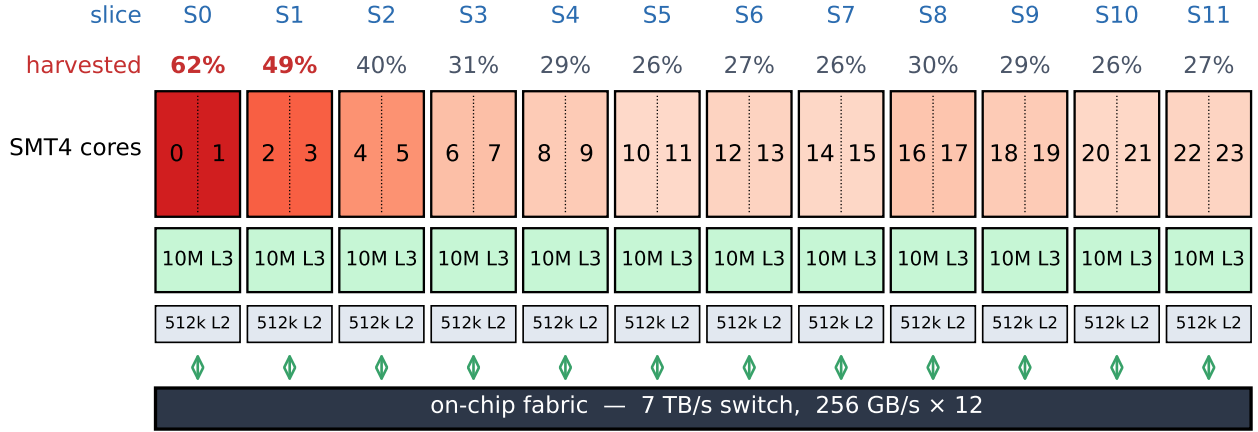


Fig. 1. POWER9 slice organisation (after [2]) annotated with the measured per-slice harvest rate across 1,960 Marconi100 sockets. Each slice holds two SMT4 cores sharing 512kB L2 and 10MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which we show is *not* established as the physical die ordering (Sec. IV-F).**

F. The slice index is not established as the physical ordering

Free classical MDS on the pooled 24×24 matrix is dominated by noise (leading axis 20.5% of variance, correlation with core index only +0.34), so we instead score explicit candidate floorplans by correlating predicted physical distance against measured thermal coupling (Fig. 5). The identity 1×12 linear arrangement scores -0.271 , better than a random ordering ($z = -2.22$, $p = 0.025$) but *worse* than a 3×4 grid (-0.565); a hill-climbing search finds an ordering scoring -0.715 whose rank correlation with the identity ordering is only +0.34.

We therefore do not claim the sensor index is monotonic in physical die position. The index-space clustering of Fig. 3 is a robust empirical fact, but its interpretation as *spatial* defect clustering is not established by our data and must be stated as a hypothesis. **[TODO: obtain an annotated POWER9 die shot or OpenBMC sensor-ID mapping to settle this.]**

G. A procurement-lot boundary at node 440

Node IDs map to position as $\text{rack} = \lfloor \text{id}/20 \rfloor$ [6]. A changepoint scan maximises Welch t at rack 22 (node 440), $t = 26.6$ (Fig. 6).

TABLE II
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY.

Population	Sockets	$P(\text{slice } 0)$	Distinct patterns
Lot A (racks 0–21)	880	88.2%	230
Lot B (racks 22–48)	1,082	39.3%	416

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.568 vs 1.454 ± 0.008 , $z = +14.1$).

H. Configuration lifetime

POWER9 exposes mechanisms that *could* alter the active set after manufacture: predictive Dynamic Processor Deallocation, persistent GARD records read by hostboot at IPL and clearable with `opal-gard`, and the field core override used for licensing [8], [9]. Operating-system core offlining is invisible to the BMC and cannot affect sensor presence. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

a) *The count is very nearly invariant:* Across all 31 months—1,671,297 socket-days over 858 days and 984 nodes—**99.985% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which *cannot* arise from slice-granular fusing and therefore proves the collector can drop individual cores; the counts above 16 are unions spanning a same-day reconfiguration. Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

b) *Configurations are near-static:* Restricting to *clean* socket-days (16 cores, each at $>90\%$ of the 4,320 samples expected at 20s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

c) *Three core-guard episodes:* The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, each with the exact signature of a field deconfiguration (Table III). In every case the socket loses *exactly one slice*—a complete $(2k, 2k+1)$ pair—runs on 14 cores for 2–4 consecutive fully-sampled days, and in at least one case the slice is subsequently restored, consistent with an operator clearing the GARD record [9]. The measured rate is **3 episodes across 1,968 sockets over 2.5 years**, i.e. 0.15%

of sockets. Durations are lower bounds, since an episode is counted only on fully-sampled days.

Ruling out a collector fault. The competing explanation is that the collector dropped exactly those two metric streams. We test it directly: for each episode we ask whether the *other* sensors on the same node—in particular on the same socket—kept reporting. They did. In all three episodes the two slice sensors read zero for the entire window while **all ten control metrics, including `pX_power` and `pX_vdd_temp` on the same socket, sustained a full 4,320 samples per day** (Fig. 9). A collection fault that silences precisely the two cores of one slice for several consecutive days, while leaving every other sensor on the same socket at full cadence, is not a plausible account. We therefore attribute these episodes to hardware deconfiguration; to our knowledge this is the first direct observation of POWER9 field core guarding in a deployed system.

TABLE III

CORE-GUARD EPISODES. BASELINES ARE THE ADJACENT OBSERVED DAY, NOT THE SOCKET’S GLOBAL MODAL SET, SINCE TWO OF THESE SOCKETS ALSO CHANGE CONFIGURATION ELSEWHERE IN THEIR LIFE.

Socket	Dates	Days	Slice lost	Restored
347 p1	2020-03-13–16	4	4 (cores 8,9)	yes
411 p1	2021-11-13–15	3	6 (cores 12,13)	yes [†]
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	not observed

[†]restored via a day that also spans a reconfiguration.

d) Changes never occur during operation: Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting (median 2d), against a background in which only 4.2% of consecutive reporting days are more than one day apart. Configuration changes are boot-time events, exactly as the guard/deconfiguration mechanism predicts.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1) and 342 (p0) appear to change within a single day, and produce the dataset’s only 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting gap*: on node 315 the outgoing cores {4,5,20,21} report until 08:42 and the incoming cores {8,9,10,11} resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, indicating a coordinated maintenance reboot. Day-granularity aggregation is therefore *conservative* here: it cannot see sub-day downtime, so our transition-gap statistics understate how consistently changes coincide with reboots.

e) Changes concentrate on a few dates: The 235 transitions involve 132 nodes on 81 distinct dates, but are heavily clustered: 2020-03-20 alone accounts for 63 and the top three dates for 41%. Such synchronised, fleet-wide changes are more consistent with an administrative or firmware event than with independent per-node hardware work (see OI-2).

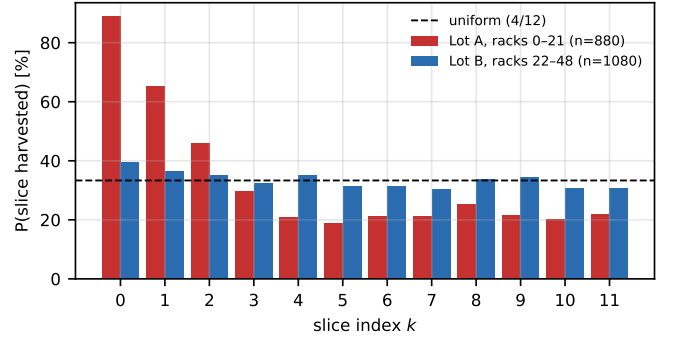


Fig. 2. Per-slice harvest probability by procurement lot.

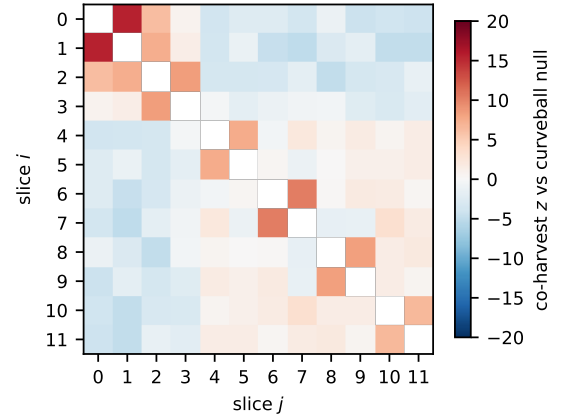


Fig. 3. Co-harvest z against the curveball null. Excess is confined to near-diagonal (index-adjacent) slice pairs.

f) Some apparent changes are socket relabelling: When node n ’s socket p_0 loses exactly the slices p_1 gains and vice versa, the dies are unchanged and only the p_0/p_1 labels have swapped. This accounts for **20%** of the 230 transitions where both sockets are observable.

g) Caveat: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. The tight clustering of 86 transitions into three days is consistent with (c) and we cannot separate these from telemetry alone. **[TODO: seek CINECA maintenance logs for March 2020.]**

I. The harvest map has no operational consequence

Recovering a hidden hardware property is only useful if it changes something. We therefore ask whether the harvest map predicts socket behaviour, using a **within-node paired contrast**: the two sockets of a node share chassis, airflow, inlet air, power supply and—to first order—workload, so the difference $p_0 - p_1$ removes the node-level confounds that make cross-node power comparisons unreliable. We summarise each socket’s map by the mean index of its active slices, their dispersion, the number of adjacent active pairs, and the count

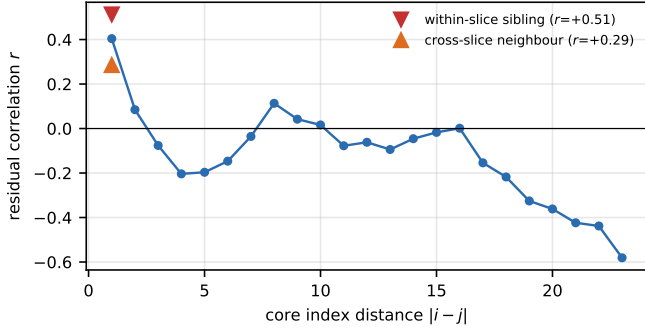


Fig. 4. Residual thermal correlation versus core index distance. The within-versus cross-slice contrast at distance 1 controls exactly for index gap.

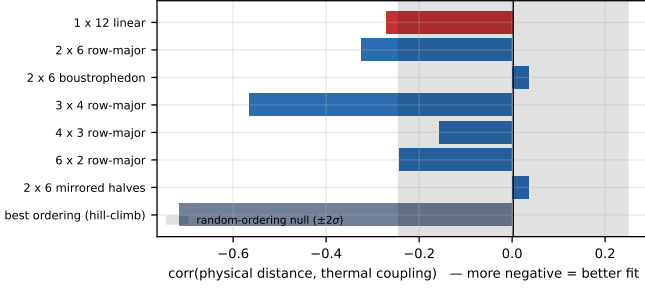


Fig. 5. Candidate floorplans scored against measured thermal coupling. The linear index ordering (red) beats a random ordering only marginally and is outperformed by a 3×4 grid.

of active slices in the low half, and correlate the socket-to-socket difference in each against the socket-to-socket difference in mean CPU power over six months (5,880 node-months, Fig. 11).

The associations are statistically detectable and practically negligible. The strongest is between the mean active-slice index and socket power: $r = +0.039$ (permutation $p = 0.003$), explaining **0.16% of variance**, with a slope of $+0.46$ W per unit of slice index (95% CI $[+0.16, +0.76]$). Across the full observed range of the predictor this is **3.1 W**, against a socket-to-socket power spread of 17.8 W. For calibration, the same paired design shows socket p0 draws 7.7 W more than socket p1 on average ($t = 44.6$)—mere physical position in the chassis matters more than twice as much as the entire range of harvest-map variation.

We therefore report a deliberate negative result: **harvest maps are operationally invisible**. Schedulers, power models and node-ranking heuristics need not represent them, and node-to-node performance variability in this system must be sought elsewhere. The result also strengthens the provenance interpretation of Sec. IV-G: because the map has no behavioural footprint, the lot boundary cannot be an artefact of operational differences between racks.

V. DISCUSSION

a) Why is slice 0 special?: The single most striking number is that slice 0 is harvested on 88.2% of Lot A sockets

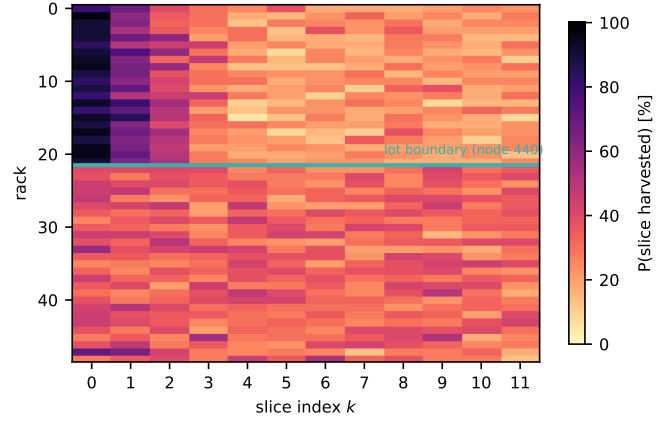


Fig. 6. Harvest probability by rack and slice. The lot boundary at rack 22 is visible as an abrupt change.

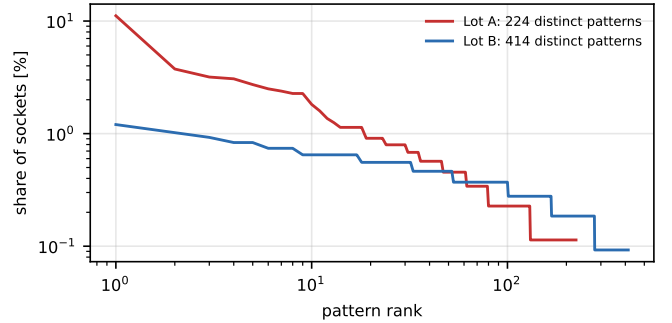


Fig. 7. Rank-frequency of disable patterns. Lot B is markedly more diverse.

but only 39.3% of Lot B. Two explanations are live. Under *defect-driven* harvesting, fusing follows wherever killer defects land; this is supported by the index-adjacency clustering of Sec. IV-D and by Lot B's nearly flat marginals, which are close to what a spatially random defect process would give. Under *policy-driven* harvesting, slice 0 abuts a shared structure and is preferentially fused for reasons of yield policy, binning recipe, or physical design; an 88% rate on one specific slice is very hard to obtain from random defects. The data are consistent with Lot A being policy- or recipe-dominated and Lot B defect-dominated, i.e. with a change of binning recipe or die source between the two procurements. We cannot settle this from one machine and report both.

b) The direction of causation is fixed.: A tempting objection to Sec. IV-G is that rack position is confounded with cooling and workload. It cannot be: harvesting is performed by fusing at manufacture, months before a part is installed, so nothing about a socket's position in the machine room can influence its harvest map. The only admissible causal direction runs from procurement batch to rack, which is precisely the provenance signal we claim.

c) What this buys an operator.: Three things. Fleet telemetry can audit a procurement without vendor cooperation, detecting whether a delivery is homogeneous. It can detect

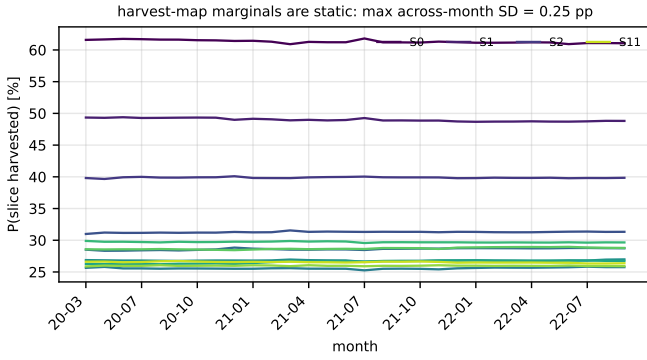


Fig. 8. Per-slice harvest rate in each of the 31 months. The largest across-month standard deviation of any slice is 0.25 percentage points.

field deconfiguration events that would otherwise be visible only in service logs. And, per Sec. IV-I, it establishes when a hardware property can safely be *ignored*.

VI. RELATED WORK

[TODO: expand; the citations below are located but unverified.] Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [10], turbo-boost variation across nominally identical processors [11], and accelerator-level variability in GPU-rich systems [12]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [13] and spatial wafer-map models [14] describe clustered killer defects but are validated on fab data unavailable to system owners. To our knowledge no prior work recovers per-die harvesting maps from deployed-system telemetry, which is the gap this paper addresses.

VII. THREATS TO VALIDITY AND OPEN ISSUES

Every unresolved item is tracked with an identifier in `OPEN_ISSUES.md` in the artefact repository, which records what is known, what would settle it, and where to look. **The items marked *blocking* below must be resolved before this paper is submitted.**

- 1) **Sensor presence $\neq$ core existence.** Mitigated by 99.985% of 1.67M socket-days reporting exactly 16 cores, agreement with the published 16-core SKU [4], the perfect slice-pair structure, and the control of Sec. IV-H showing that when core sensors do vanish, every other sensor on the socket keeps reporting.
- 2) **[OI-1, *blocking*] The sensor index may not be the physical die position** (Sec. IV-F). The spatial reading of the clustering in Sec. IV-D is a hypothesis, not a result. Needs an annotated die shot or the hostboot sensor-ID mapping.
- 3) **[OI-2, *blocking*] A March 2020 BMC firmware update could have remapped sensor IDs**, mimicking reconfiguration; 86 of the 235 transitions fall in the three



Fig. 9. The three core-guard episodes. Grey = harvested at manufacture, blue = configured and reporting, red = configured but silent. In every case exactly one slice goes silent while all ten control sensors on the same node stay at full cadence.

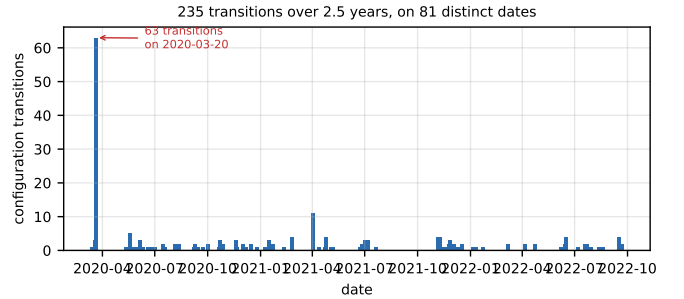


Fig. 10. Configuration transitions over the full record. 2020-03-20 alone accounts for 63.

days 2020-03-19–21. Needs CINECA maintenance logs. Does not affect the boot-time-only conclusion.

- 4) **[OI-4, *blocking*] Five references are unverified** and a novelty check against prior telemetry-based binning inference is outstanding.
- 5) **[OI-3] Defect-driven versus policy-driven harvesting** cannot be separated from one machine; both are reported.
- 6) **[OI-5] Collection dropout may not be random.** Monthly coverage ranges from 18% to 99.4%; if dropout correlates with lot, Sec. IV-G needs a robustness pass.
- 7) **[OI-6, OI-7] both resolved.** The sub-16 socket-days were dominated by an extraction defect on our side, now gated; the nine that survive are the guard episodes of Sec. IV-H. The two nodes appearing to change within a day are same-day reboots.
- 8) **[OI-8] The mechanism behind p0/p1 relabelling is unknown;** any per-socket longitudinal study on this dataset risks silently mixing two dies.

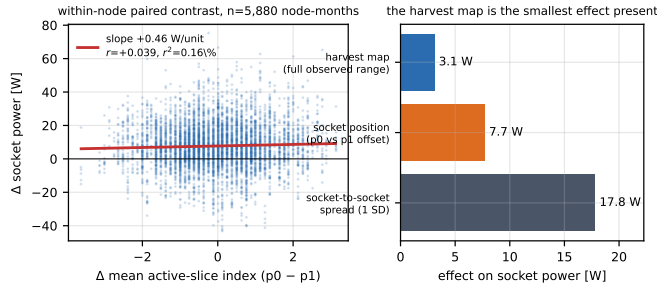


Fig. 11. Operational impact of the harvest map. Left: within-node paired contrast. Right: the harvest map’s full observed range moves socket power less than half as much as mere socket position, and a sixth as much as the socket-to-socket spread.

- 9) **Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts; the pooled estimator may be improvable [OI-11].
- 10) **[OI-10] Single system**; findings may reflect CINECA procurement rather than POWER9 generally.

VIII. CONCLUSION

Out-of-band telemetry that supercomputer operators already collect inadvertently publishes a manufacturing secret: which units a vendor fused off each individual die. Recovering these maps for 1,962 POWER9 sockets shows harvesting operates exactly at the slice—the unit sharing an L2/L3 block—with no exceptions in 1.67M socket-days; that the maps are per-die rather than per-SKU, with 443 of 495 possible patterns realised; that harvested slices cluster in sensor-index space; and that a sharp changepoint at node 440 separates two procurement populations with different binning statistics. Over 2.5 years the maps are static to within 0.25 percentage points per month, change only across reboots, and exhibit three field core-guard episodes, each removing exactly one slice while every other sensor on the socket keeps reporting.

Two boundaries deserve emphasis. We do *not* establish that the sensor index tracks physical die position, so the clustering result is stated in index space and its spatial reading remains a hypothesis. And the harvest map turns out to have no measurable operational effect—a negative result we consider as useful as the positive ones, since it tells operators what they can ignore. The broader point is that fleet telemetry is a viable instrument for hardware provenance, and that its unintended disclosures deserve attention from those who publish such datasets.

ARTEFACT AVAILABILITY

All analysis code, the derived per-(node, socket, core, day) tables, and the recovered harvest maps are available at the artefact repository accompanying this paper. Every result is reproducible from the public M100 ExaData Zenodo releases [1]; the derived tables (15 MB) suffice for all structural and longitudinal results without re-downloading the 49.9 TB source.

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